Assignment 1

1 Signal norms

Definition 1. *Given a signal* $r: \mathbb{R}^{\geq 0} \to \mathbb{R}^n$*, the* ***p-norm*** *of* $r(t)$*, for* $p \geq 1$*, is defined by*

$$\|r(t)\|_p \;=\; \left(\int_0^\infty \sum_{i=1}^{n} |r_i(\tau)|^p d\tau \right)^{\frac{1}{p}}$$

Definition 2. *Given a signal* $r: \mathbb{R}^{\geq 0} \to \mathbb{R}^n$*, the* $\boldsymbol{\infty}$***-norm*** *of* $r(t)$ *is defined by*

$$\|r(t)\|_\infty \;=\; \max_i \sup_t |r(t)|_i$$

Definition 3. *A signal* $r(t)$ *has finite p-norm* $r \in L_p$ *if* $\|r(t)\|_p < \infty$.

Theorem 4. *If* $r \in L_1 \cap L_\infty \Rightarrow r \in L_2$.

Proof.

We have

$$\begin{aligned}
\|r(t)\|_2^2 \;&=\; \int_0^\infty \sum_{i=1}^{n} |r_i(\tau)|^2 d\tau \\
&=\; \int_0^\infty \sum_{i=1}^{n} \left(|r_i(\tau)|\, |r_i(\tau)| \right) d\tau \\
&\leqslant\; \int_0^\infty \sum_{i=1}^{n} \left(|r_i(\tau)| \cdot \sup_t |r_i(t)| \right) d\tau \\
&\leq\; \int_0^\infty \sum_{i=1}^{n} \left(|r_i(\tau)| \cdot \max_i \sup_t |r_i(t)| \right) d\tau \\
&=\; \max_i \sup_t |r_i(t)| \cdot \int_0^\infty \sum_{i=1}^{n} |r_i(\tau)| d\tau \\
&=\; \|r(t)\|_\infty \|r(t)\|_1
\end{aligned}$$

Since $\|r(t)\|_\infty < \infty$ and $\|r(t)\|_1 < \infty$ by assumption, the result follows.

□

2 Lyapunov stability theorem and Barbalat's lemma

Definition 5. *For a dynamical system $\dot{x} = f(x(t))$, an equilibrium point x_e is* ***stable*** *if a perturbation of the initial state x_0 keeps the system within a prescribed distance of x_e, and that distance can be chosen arbitrarily small.*

$$\forall \varepsilon > 0 \quad \exists \delta_\varepsilon > 0 \quad / \quad \|x_0 - x_e\| < \delta_\varepsilon \quad \Rightarrow \quad \|\varphi(t, x_0) - x_e\| < \varepsilon \quad \forall t \geqslant 0$$

Theorem 6. ***(Lyapunov stability theorem)***

Consider the autonomous system $\dot{x} = f(x)$ with equilibrium $x_e = 0$. If $\exists\ V \in C^1$, $V>0$, such that $\dot{V} \leq 0$, then the equilibrium $x_e = 0$ is stable.

Theorem 7. ***(Barbalat's lemma)***

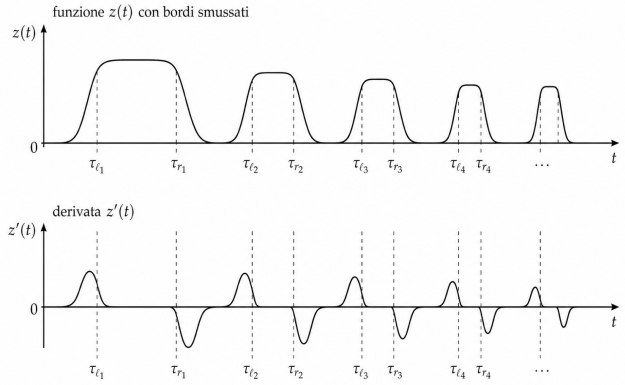
If a function $r(t) \in L_2 \cap L_\infty$ has a bounded derivative ($\dot{r}(t) \in L_\infty$), then it converges asymptotically to zero.

Remark 8.

- The assumption $r \in L_\infty$ is stronger than necessary.
- The assumption $r(t) \in L_2 \bigcap L_\infty$ can be replaced by $r(t) \in L_p \bigcap L_\infty$, where $p \geq 1$.
- The bounded-derivative condition $\dot{r}(t) \in L_\infty$ matters. For example, consider a function consisting of rectangular pulses with areas $1, \frac{1}{2}, \frac{1}{4} \ldots$, spaced one second apart. The total integral is

$$\begin{aligned} \int_0^\infty f(t)\mathrm{dt} &= \sum_{k=0}^{\infty} \frac{1}{2^k} \\ &= \frac{1}{1-\frac{1}{2}} \\ &= 2 \end{aligned}$$

  Hence $r \in L_1$. Since the signal is also bounded, we obtain $r \in L_1 \cap L_\infty, r \in L_2$.



  To examine Barbalat's lemma, smooth the edges of the pulses while keeping their areas unchanged. The function rises and falls over progressively shorter intervals, so the magnitude of its derivative grows without bound. Thus, if $r(t) \in L_2 \bigcap L_\infty$ but $\dot{r}(t) \notin L_\infty$, the conclusion $\lim_{t\to\infty} r(t) = 0$ is not guaranteed.

3 Passive systems

Definition 9. *A square system is* ***passive*** *if it is dissipative with respect to the supply rate* $s(u, y) = u^T y$*, meaning that there exists a storage function* $S(x(t)) \geq 0$ *satisfying*

$$S(x(t_1)) \;\leq\; S(x(t_0)) + \int_{t_0}^{t_1} u^T(t) y(t) \cdot \mathrm{dt}$$

For $S \in C^1$*, the system is passive if*

$$S' \;\leq\; u^T y$$

Remark 10. In an electrical system with voltages as inputs and currents as outputs, the passivity supply rate represents the total power. In a mechanical system with torques as inputs and moments as outputs, the supply rate represents mechanical power.

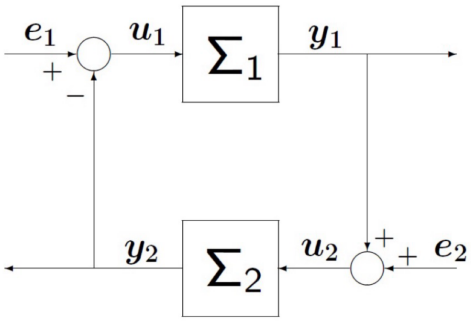
4 Passivity theorem

Theorem 11. *Consider the following nonlinear systems*

$$\Sigma_1 \quad \begin{array}{l} \dot{x_1} = f_1(x_1, u_1) \\ y_1 = h_1(x_1, u_1) \end{array} \qquad \Sigma_2 \quad \begin{array}{l} \dot{x_2} = f_2(x_2, u_2) \\ y_2 = h_2(x_2, u_2) \end{array}$$

with input signals u_1 and u_2 satisfying

$$u_1 = e_1 - y_2 \qquad u_2 = e_2 + y_1$$



1. *If Σ_1 and Σ_2 are passive with storage functions S_1 and S_2, then the resulting system is passive with storage function $S = S_1 + S_2$.*

Proof.

The block diagram gives the input of the interconnected system as

$$e = \begin{pmatrix} e_1 \\ e_2 \end{pmatrix}$$

the state as

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

and the output as

$$y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

By definition,

$$\begin{aligned} S_1(x_1(t_1)) &\leq S_1(x_1(t_0)) + \int_{t_0}^{t_1} u_1^T(\tau) y_1(\tau) \quad d\tau \\ S_2(x_2(t_1)) &\leq S_2(x_2(t_0)) + \int_{t_0}^{t_1} u_2^T(\tau) y_2(\tau) \quad d\tau \end{aligned}$$

or equivalently,

$$\begin{aligned} S_1(x_1(t_1)) &\leq S_1(x_1(t_0)) + \int_{t_0}^{t_1} (e_1(\tau) - y_2(\tau))^T y_1(\tau) \quad d\tau \\ S_2(x_2(t_1)) &\leq S_2(x_2(t_0)) + \int_{t_0}^{t_1} (e_2(\tau) + y_1(\tau))^T y_2(\tau) \quad d\tau \end{aligned}$$

Adding the two inequalities gives

$$\begin{aligned}
S_1(x_1(t_1)) + S_2(x_2(t_1)) &\leq S_1(x_1(t_0)) + \int_{t_0}^{t_1} (e_1(\tau) - y_2(\tau))^T y_1(\tau) \quad d\tau \\
&\quad + S_2(x_2(t_0)) + \int_{t_0}^{t_1} (e_2(\tau) + y_1(\tau))^T y_2(\tau) \quad d\tau \\
&= S_1(x_1(t_0)) + S_2(x_2(t_0)) + \int_{t_0}^{t_1} e_1^T(\tau) y_1(\tau) - y_2^T(\tau) y_1(\tau) \\
&\quad + e_2^T(\tau) y_2(\tau) + y_1^T(\tau) y_2(\tau) \quad d\tau \\
&= S_1(x_1(t_0)) + S_2(x_2(t_0)) + \int_{t_0}^{t_1} e_1^T(\tau) y_1(\tau) + e_2^T(\tau) y_2(\tau) \qquad d\tau \\
&= S_1(x_1(t_0)) + S_2(x_2(t_0)) + \int_{t_0}^{t_1} \begin{pmatrix} e_1^T(\tau) & e_2^T(\tau) \end{pmatrix} \begin{pmatrix} y_1(\tau) \\ y_2(\tau) \end{pmatrix} \qquad d\tau \\
&= S_1(x_1(t_0)) + S_2(x_2(t_0)) + \int_{t_0}^{t_1} e^T(\tau) y(\tau) \qquad d\tau
\end{aligned}$$

Setting $S(x) = S_1(x_1) + S_2(x_2)$ gives

$$S(x(t_1)) \leq S(x(t_0)) + \int_{t_0}^{t_1} e^T(\tau) y(\tau) \quad d\tau$$

This proves the result.

□

5 Scalar parameter estimation

Theorem 12.

Consider the scalar system

$$\dot{x} \;=\; -a\,x + b\,u$$

where $a > 0$ and b are unknown constant parameters. Define

- *$\hat{x}$ as the state estimate*
- *$\hat{a}$ as the estimate of parameter a*
- *$\hat{b}$ as the estimate of parameter b*
- *a_m as a design parameter*
- *$e = \hat{x} - x$ as the state estimation error*

Let the parallel model be defined by

$$\begin{array}{lrcl} (P) & \dot{\hat{x}} & = & -\hat{a}\,\hat{x} + \hat{b}\,u \\ & \dot{\hat{a}} & = & e\,\hat{x} \\ & \dot{\hat{b}} & = & -e\,u \end{array}$$

and the series-parallel model by

$$\begin{array}{lrcl} (\mathrm{SP}) & \dot{\hat{x}} & = & -a_m\,\hat{x} + (a_m - \hat{a})x + \hat{b}\,u \\ & \dot{\hat{a}} & = & e\,x \\ & \dot{\hat{b}} & = & -e\,u \end{array}$$

If the actual system is stable and the input is bounded, the state estimation error converges asymptotically to zero.

$$\lim_{t\to\infty} e(t) \;=\; 0$$

Proof. (**parallel model**)

We first prove the theorem for the parallel model.

We begin by deriving the equations that define the parallel model.

Define

- $e = \hat{x} - x$ as the state estimation error
- $\tilde{a} = \hat{a} - a$ as the estimation error for parameter a
- $\tilde{b} = \hat{b} - b$ as the estimation error for parameter b

If

$$\dot{\hat{x}} \;=\; -\hat{a}\hat{x} + \hat{b}u$$

the state estimation error dynamics for input $u_1 = \begin{pmatrix} \tilde{a} \\ \tilde{b} \end{pmatrix}$ are given by

$$\begin{aligned}
\dot{e} &= \dot{\hat{x}} - \dot{x} \\
&= -\hat{a}\hat{x} + \hat{b}\,u - (-a\,x + b\,u) \\
&= -\hat{a}\hat{x} + \hat{b}\,u + a\,x - b\,u \\
&= -\hat{a}\hat{x} + a\,x + (\hat{b} - b)\,u \\
&= -\hat{a}\hat{x} + a(\hat{x} - e) + \tilde{b}u \\
&= -a\,e - (\hat{a} - a)\hat{x} + \tilde{b}u \\
&= -a\,e - \tilde{a}\hat{x} + \tilde{b}u \\
&= -a\,e + \begin{pmatrix} -\hat{x} & u \end{pmatrix}\begin{pmatrix} \tilde{a} \\ \tilde{b} \end{pmatrix}
\end{aligned}$$

If $S_1 = \frac{1}{2}e^2$, then

$$\begin{aligned}
\dot{S}_1 &= e\,\dot{e} \\
&= e\left(-a\,e + \begin{pmatrix} -\hat{x} & u \end{pmatrix}\begin{pmatrix} \tilde{a} \\ \tilde{b} \end{pmatrix}\right) \\
&= -a\,e^2 + e\begin{pmatrix} -\hat{x} & u \end{pmatrix}\begin{pmatrix} \tilde{a} \\ \tilde{b} \end{pmatrix} \\
&= -a\,e^2 + \begin{pmatrix} -\hat{x}\,e & u\,e \end{pmatrix}\begin{pmatrix} \tilde{a} \\ \tilde{b} \end{pmatrix}
\end{aligned}$$

The parallel system is therefore passive with input

$$u_1 \;=\; \begin{pmatrix} \tilde{a} \\ \tilde{b} \end{pmatrix}$$

and output

$$y_1 \;=\; \begin{pmatrix} -\hat{x}\,e \\ u\,e \end{pmatrix}$$

Since the derivative of the storage function contains the terms

$$-\tilde{a}\;\hat{x}\,e + \tilde{b}\;u\,e$$

the storage function need not decrease, so it cannot be used directly as a Lyapunov function.

To cancel these terms, introduce a second passive system Σ_2 whose storage function depends on the parameter estimation errors.

$$S_2 = \frac{1}{2}\tilde{a}^2 + \frac{1}{2}\tilde{b}^2.$$

Since

$$\dot{S}_2 = \tilde{a}\ \dot{\hat{a}} + \tilde{b}\ \dot{\tilde{b}}.$$

to obtain the following derivative for the total storage function

$$\dot{S} = \dot{S}_1 + \dot{S}_2 = -a\, e^2,$$

the derivative of the second storage function must satisfy

$$\dot{S}_2 = \tilde{a}\ \hat{x}\, e - \tilde{b}\ u\, e.$$

or equivalently,

$$\begin{aligned} \dot{\hat{a}} &= \hat{x}\, e \\ \dot{\tilde{b}} &= -u\, e \end{aligned}$$

Since

$$\begin{aligned} \tilde{a} &= \hat{a} - a \\ \tilde{b} &= \hat{b} - b \end{aligned}$$

and a and b are constant, we obtain

$$\begin{aligned} \dot{\hat{a}} &= \hat{x}\, e \\ \dot{\tilde{b}} &= -u\, e \end{aligned}$$

In this case, the total storage function

$$\begin{aligned} \dot{S} &= -a\, e^2 \\ &\leq 0 \end{aligned}$$

It follows that

$$S(x(t)) \leq S(x(0)) \qquad \forall t \geq 0$$

Recalling that

$$S(x(t)) = \frac{1}{2}e^2 + \frac{1}{2}\tilde{a}^2 + \frac{1}{2}\tilde{b}^2$$

we obtain $e, \tilde{a}, \tilde{b} \in L_\infty$.

We also have

$$\begin{aligned} S(x(t)) - S(x(0)) &= \int_0^t \dot{S}(t)\, \mathrm{dt} \\ &= \int_0^t -a\, e^2(t) \mathrm{dt} \end{aligned}$$

Since $S(x(t)) \geq 0 \quad \forall t \geq 0$

$$\begin{aligned} \int_0^t -a\, e^2(t)\mathrm{dt} &= S(x(t)) - S(x(0)) \\ &\geq -S(x(0)) \\ \int_0^t e^2(t)\mathrm{dt} &\leq \frac{S(x(0))}{a} \\ &< \infty \end{aligned}$$

For $t \to \infty$

$$\begin{aligned} \|e\|_2^2 &= \int_0^\infty e^2(t)\mathrm{dt} \\ &\leq \frac{S(x(0))}{a} \\ &< \infty \end{aligned}$$

Equivalently, $e \in L_2$.

Finally, since

$$\dot{e} = -a\, e - \tilde{a}\hat{x} + \tilde{b}u$$

and

- e, $\tilde{a}$ and $\tilde{b}$ are bounded
- u is bounded by assumption
- the actual system is stable, so x is bounded
- $\hat{x} = x + e$

we obtain $\dot{e} \in L_\infty$. By Barbalat's lemma,

$$\lim_{t \to \infty} e(t) = 0$$

□

Proof. (**series-parallel model**)

We now prove the theorem for the series-parallel model. The reasoning follows the same steps as in the parallel case.

We begin by deriving the equations that define the series-parallel model.

Define

- $e = \hat{x} - x$ as the state estimation error
- $\tilde{a} = \hat{a} - a$ as the estimation error for parameter a
- $\tilde{b} = \hat{b} - b$ as the estimation error for parameter b

If

$$\dot{\hat{x}} \;=\; -a_m\hat{x} + (a_m - \hat{a})x + \hat{b}u$$

the state estimation error dynamics for input $u_1 = \begin{pmatrix} \tilde{a} \\ \tilde{b} \end{pmatrix}$ are given by

$$\begin{aligned}
\dot{e} &= \dot{\hat{x}} - \dot{x} \\
&= -a_m\hat{x} + (a_m - \hat{a})x + \hat{b}u - (-a\,x + b\,u) \\
&= -a_m\hat{x} + a_m x - \hat{a}x + \hat{b}u + a\,x - b\,u \\
&= -a_m\hat{x} + a_m x - (\hat{a} - a)x + (\hat{b} - b)u \\
&= -\cancel{a_m}\hat{x} + \cancel{a_m}\hat{x} - a_m e - \tilde{a}x + \tilde{b}u \\
&= -a_m e - \tilde{a}x + \tilde{b}u \\
&= -a_m e + \begin{pmatrix} -x & u \end{pmatrix}\begin{pmatrix} \tilde{a} \\ \tilde{b} \end{pmatrix}
\end{aligned}$$

If $S_1 = \frac{1}{2}e^2$, then

$$\begin{aligned}
\dot{S}_1 &= e\,\dot{e} \\
&= e\left(-a_m e + \begin{pmatrix} -x & u \end{pmatrix}\begin{pmatrix} \tilde{a} \\ \tilde{b} \end{pmatrix}\right) \\
&= -a_m\,e^2 + \begin{pmatrix} -e\,x & e\,u \end{pmatrix}\begin{pmatrix} \tilde{a} \\ \tilde{b} \end{pmatrix}
\end{aligned}$$

The series-parallel system is therefore passive with input

$$u_1 \;=\; \begin{pmatrix} \tilde{a} \\ \tilde{b} \end{pmatrix}$$

and output

$$y_1 \;=\; \begin{pmatrix} -e\,x \\ e\,u \end{pmatrix}$$

Since the derivative of the storage function contains the terms

$$-\tilde{a}\;e\,x + \tilde{b}\;e\,u$$

the storage function need not decrease, so it cannot be used directly as a Lyapunov function.

To cancel these terms, introduce a second passive system Σ_2 whose storage function depends on the parameter estimation errors.

$$S_2 = \frac{1}{2}\tilde{a}^2 + \frac{1}{2}\tilde{b}^2.$$

Since

$$\dot{S}_2 = \tilde{a}\;\dot{\tilde{a}} + \tilde{b}\;\dot{\tilde{b}}.$$

to obtain the following derivative for the total storage function

$$\dot{S} = \dot{S}_1 + \dot{S}_2 = -a\,e^2,$$

the derivative of the second storage function must satisfy

$$\dot{S}_2 = \tilde{a}\ e\,x - \tilde{b}\ e\,u$$

or equivalently,

$$\begin{aligned}\dot{\tilde{a}} &= e\,x \\ \dot{\tilde{b}} &= -e\,u\end{aligned}$$

Since

$$\begin{aligned}\tilde{a} &= \hat{a} - a \\ \tilde{b} &= \hat{b} - b\end{aligned}$$

and a and b are constant, we obtain

$$\begin{aligned}\dot{\hat{a}} &= e\,x \\ \dot{\hat{b}} &= -e\,u\end{aligned}$$

In this case, the total storage function

$$\begin{aligned}\dot{S} &= -a\,e^2 \\ &\leq 0\end{aligned}$$

The remainder of the proof follows the parallel case.

□

Remark 13.
If

$$\lim_{t\to\infty} e(t) = 0$$

we obtain

$$\begin{aligned}\lim_{t\to\infty} \dot{\tilde{a}} &= \lim_{t\to\infty} e\,\hat{x} \\ &= 0\end{aligned}$$

and

$$\begin{aligned}\lim_{t\to\infty} \dot{\tilde{b}} &= \lim_{t\to\infty} -e\,u \\ &= 0\end{aligned}$$

The parameter estimation errors $\tilde{a}$ and $\tilde{b}$ may still fail to converge to zero.

5.1 Persistently exciting signals

Definition 14. *A signal $u(t)\colon \mathbb{R}^{\geq 0} \to \mathbb{R}^m$ is **persistently exciting** if there exist three constants $t_0, T, \alpha > 0$ such that*

$$\frac{1}{T}\int_t^{t+T} u(\tau)\cdot u^T(\tau)d\tau \geq \alpha I > 0 \qquad \forall t \geq t_0$$

Theorem 15. *If the signal is persistently exciting, the parameter estimates of the parallel and series-parallel models converge to the true values of a and b.*

6 Assignment

Consider the dynamical system

$$\dot{x} = -a\,x + b\,u.$$

Assume the true parameter values are $a = 1$ and $b = 2$. Compare the performance of the parallel and series-parallel models for different values of a_m using noise-free state measurements. Then analyse the behaviour of both models in the presence of measurement noise.

Next, consider a time-varying parameter a governed by

$$a(t) = 1 + 0.1\sin\left(\frac{2\pi t}{24 \cdot 3600}\right).$$

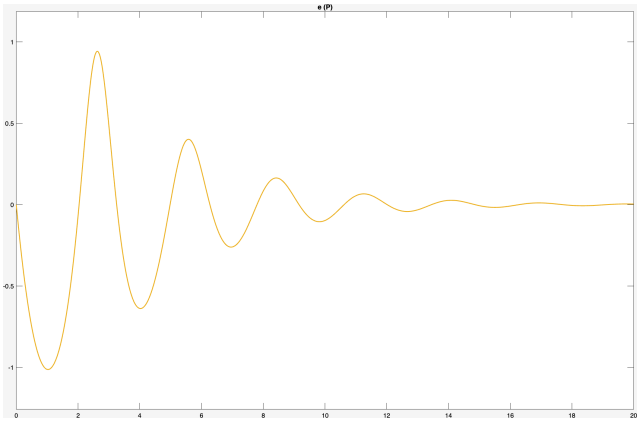
Compare the two models in terms of estimation error convergence and how well the parameter estimates track the time-varying true parameters.

7 Parameter $a = 1$, constant input $u = 1$

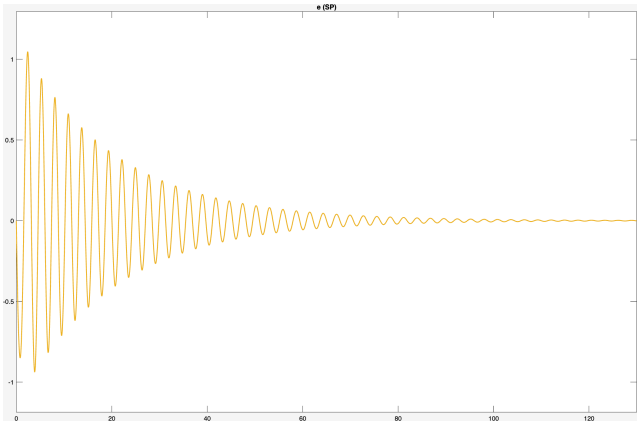
7.0.1 State estimation error $e = \hat{x} - x$

The results are as follows.

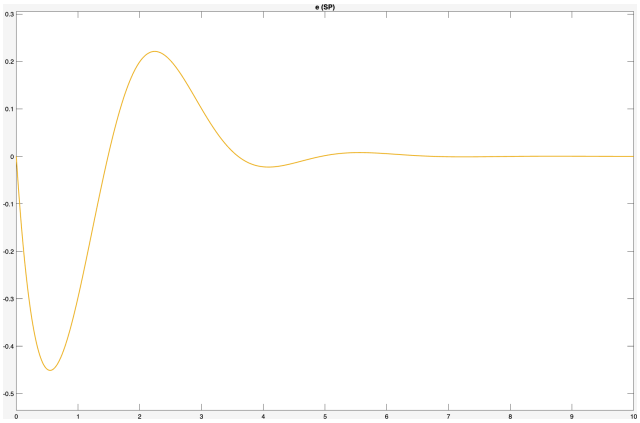
- In the parallel model, the estimation error e converges in approximately $18s$.



- In the series-parallel model with $a_m = 0.1$, the estimation error e converges in approximately $130s$.



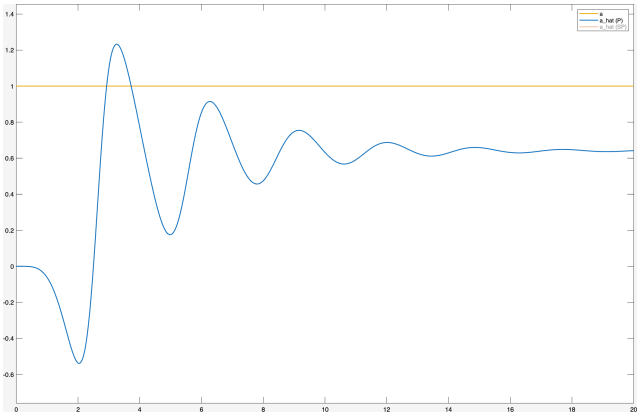
- In the series-parallel model with $a_m = 2$, the estimation error e converges in approximately $7s$.



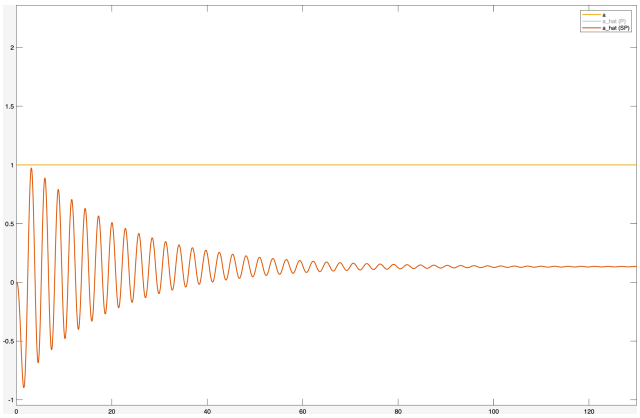
7.0.2 Estimation error for a

The results are as follows.

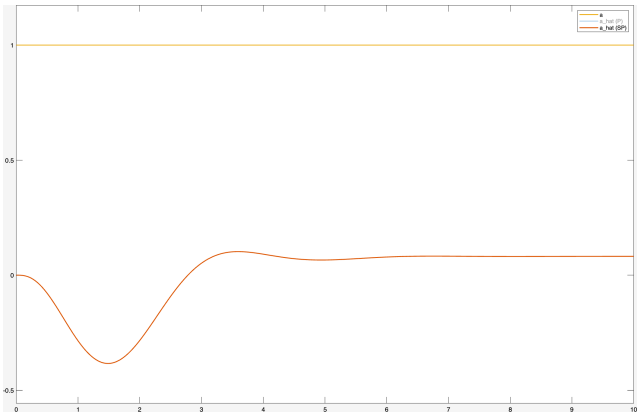
- In the parallel model, the estimation error for $a\,\hat{a}$ converges in approximately $19s$ to 0.64.



- In the series-parallel model with $a_m = 0.1$, the estimation error for a $\hat{a}$ converges in approximately $120s$ to 0.13.



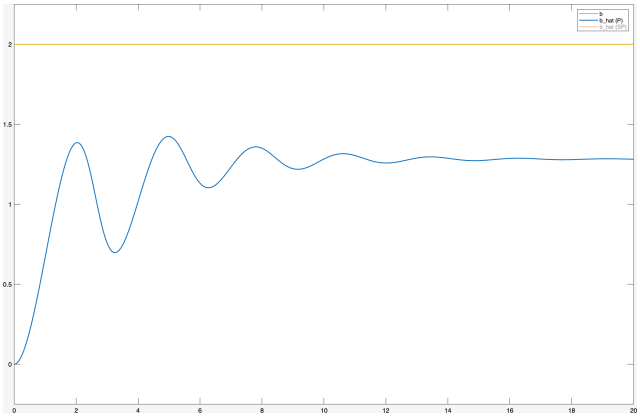
- In the series-parallel model with $a_m = 2$, the estimation error for a $\hat{a}$ converges in approximately $7s$ to 0.1.



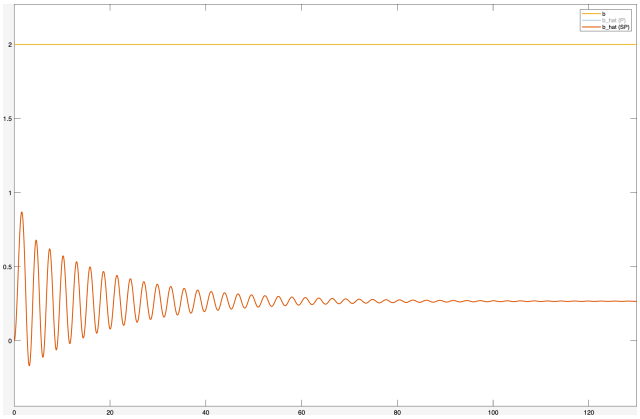
7.0.3 Estimation error for b

The results are as follows.

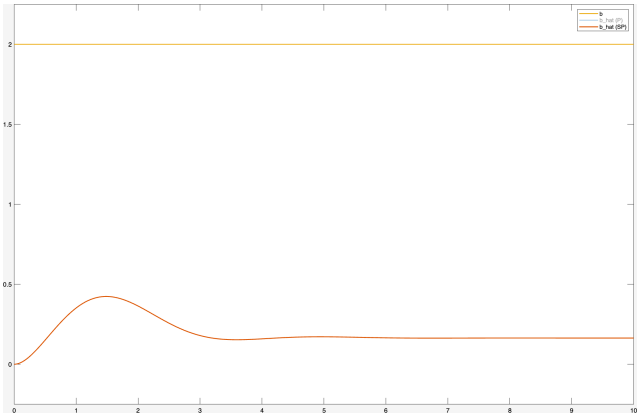
- In the parallel model, the estimation error for $b\,\hat{b}$ converges in approximately $19\,s$ to 1.28.



- In the series-parallel model with $a_m = 0.1$, the estimation error for $b\,\hat{b}$ converges in approximately $120s$ to 0.26.



- In the series-parallel model with $a_m = 2$, the estimation error for $b\,\hat{b}$ converges in approximately $6s$ to 0.16.

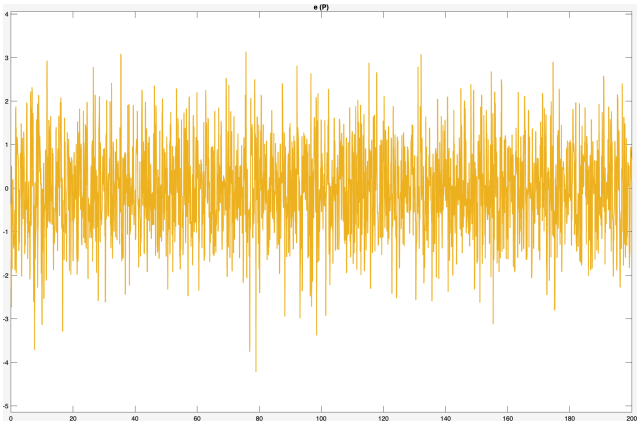


7.1 Measurement noise

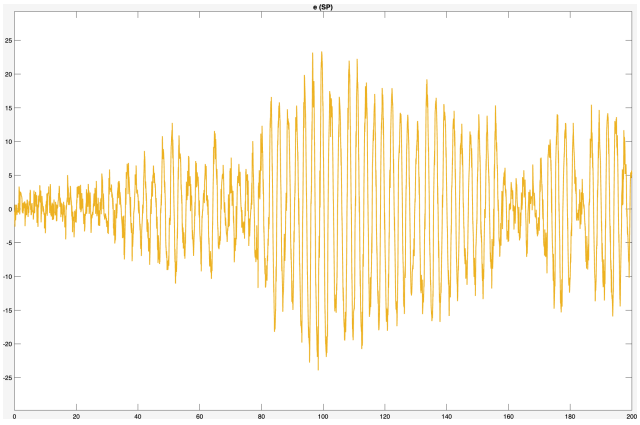
7.1.1 State estimation error $e = \hat{x} - x$

With noise in the state measurement x, the results are as follows.

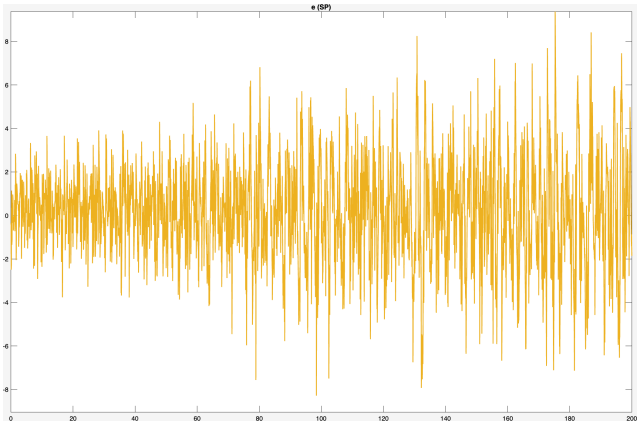
- In the parallel model, the estimation error e fluctuates between -4.5 and $+3.5$.



- In the series-parallel model with $a_m = 0.1$, the estimation error e fluctuates within ± 25.



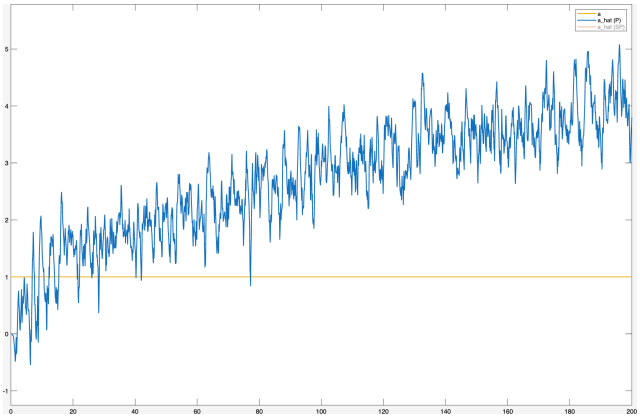
- In the series-parallel model with $a_m = 2$, the estimation error e fluctuates between -9 and $+9$.



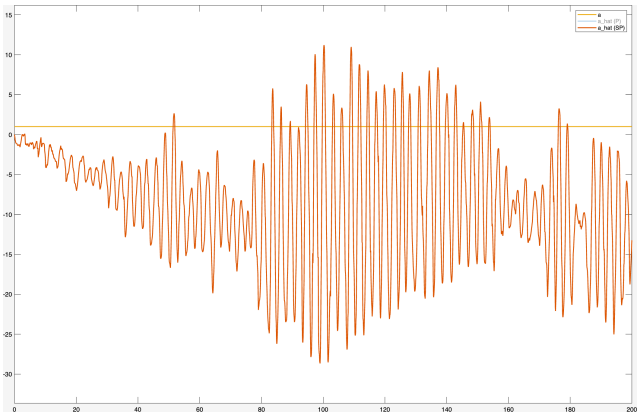
7.1.2 Estimation error for a

With noise in the state measurement x, the results are as follows.

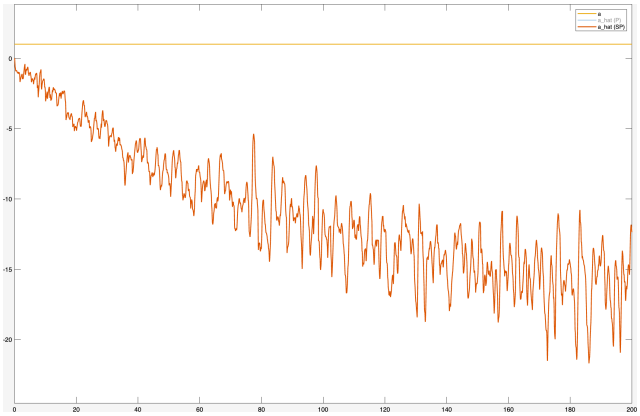
- In the parallel model, the estimation error for $a\,\hat{a}$ diverges.



- In the series-parallel model with $a_m = 0.1$, the estimation error for $a\ \hat{a}$ lies between -35 and $+15$.



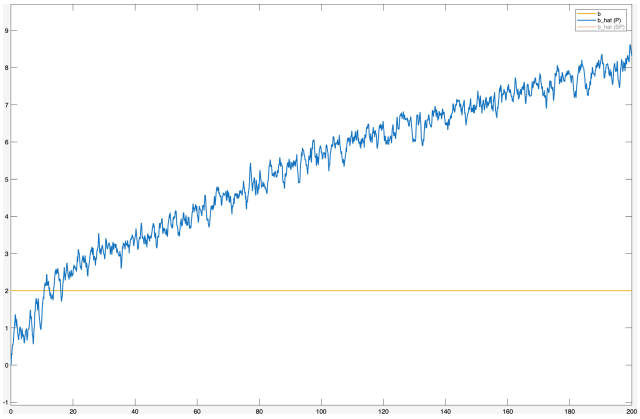
- In the series-parallel model with $a_m = 2$, the estimation error for $a\ \hat{a}$ diverges.



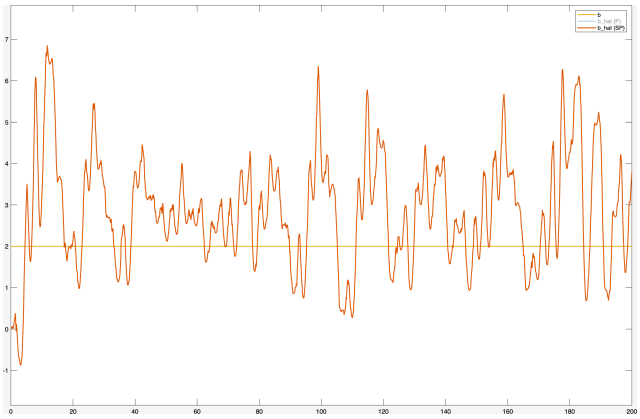
7.1.3 Estimation error for b

With noise in the state measurement x, the results are as follows.

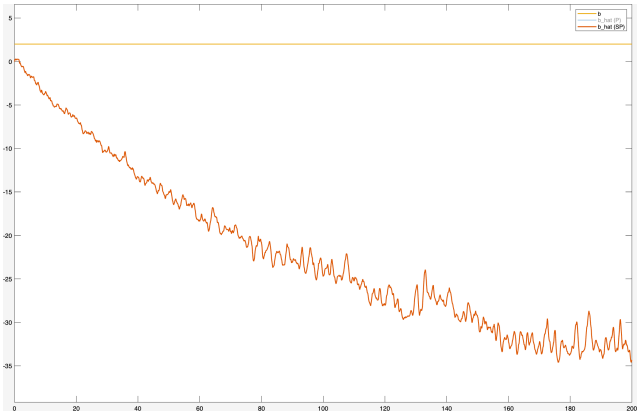
- In the parallel model, the estimation error for $b\,\hat{b}$ diverges.



- In the series-parallel model with $a_m = 0.1$, the estimation error for $b\ \hat{b}$ fluctuates between -1 and +7.



- In the series-parallel model with $a_m = 2$, the estimation error for $b\ \hat{b}$ diverges.

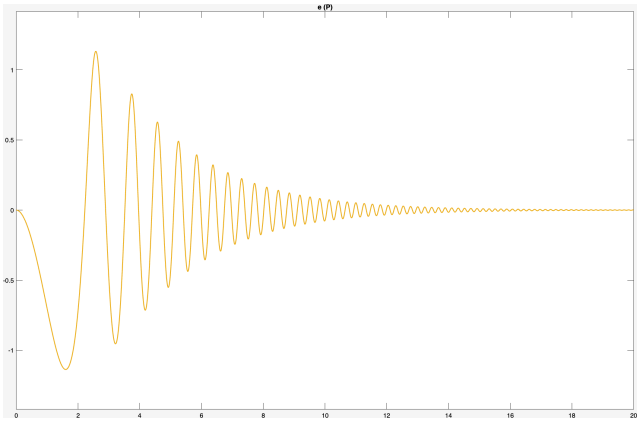


8 Parameter $a = 1$, ramp input $u = t$

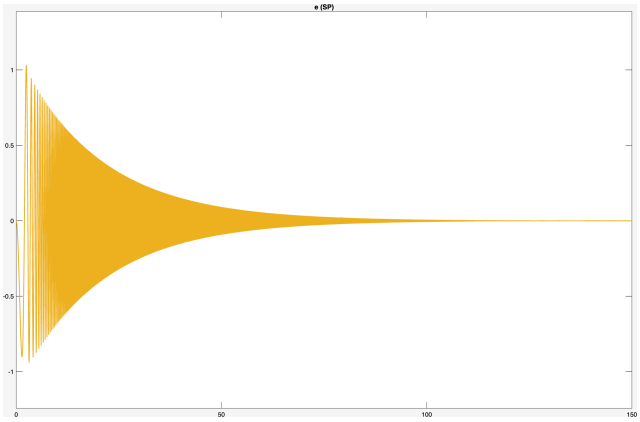
8.0.1 State estimation error $e = \hat{x} - x$

The results are as follows.

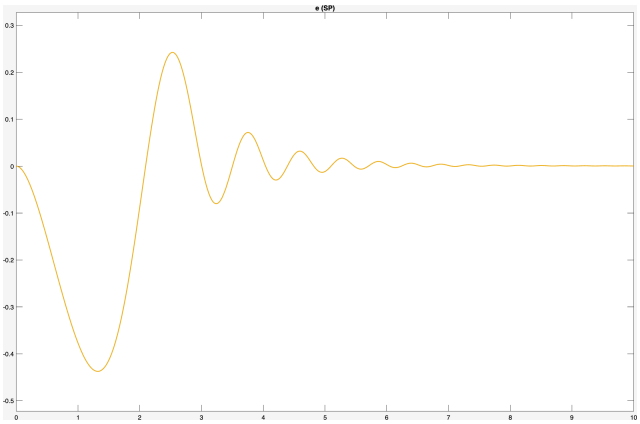
- In the parallel model, the estimation error e converges in approximately $17s$.



- In the series-parallel model with $a_m = 0.1$, the estimation error e converges in approximately $130s$.



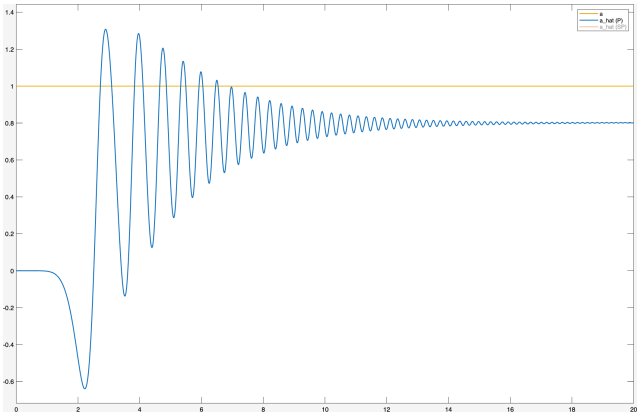
- In the series-parallel model with $a_m = 2$, the estimation error e converges in approximately $8s$.



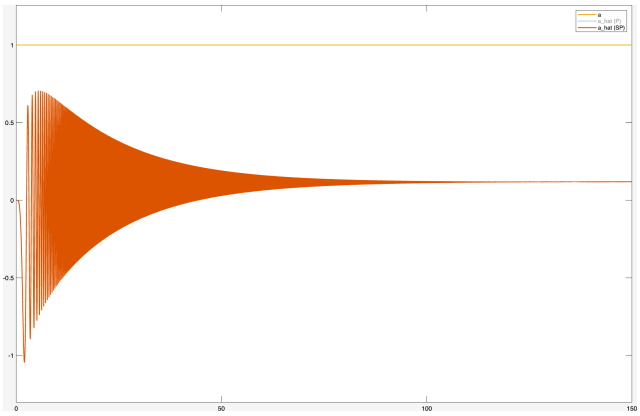
8.0.2 Estimation error for a

The results are as follows.

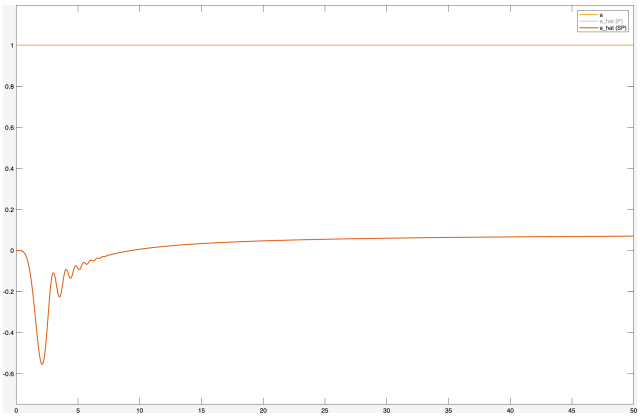
- In the parallel model, the estimation error for a $\hat{a}$ converges in approximately $18s$ to 0.8.



- In the series-parallel model with $a_m = 0.1$, the estimation error for a $\hat{a}$ converges in approximately $145s$ to 0.12.



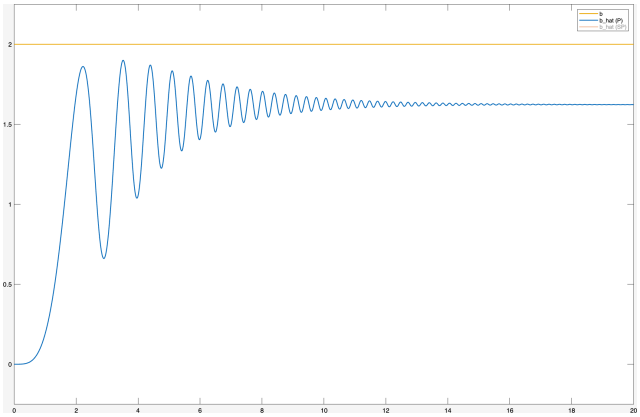
- In the series-parallel model with $a_m = 2$, the estimation error for a $\hat{a}$ converges in approximately $50s$ to 0.07.



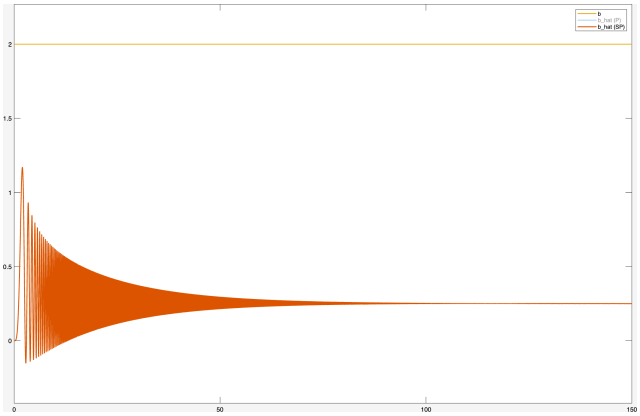
8.0.3 Estimation error for b

The results are as follows.

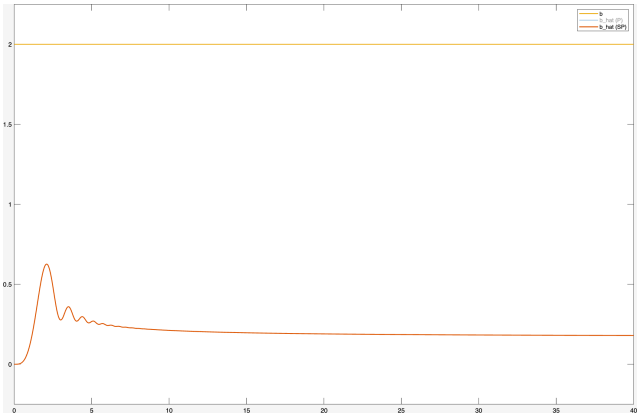
- In the parallel model, the estimation error for b $\hat{b}$ converges in approximately $18s$ to 1.62.



- In the series-parallel model with $a_m = 0.1$, the estimation error for b $\hat{b}$ converges in approximately $120s$ to 0.25.



- In the series-parallel model with $a_m = 2$, the estimation error for b $\hat{b}$ converges in approximately $30s$ to 0.18.

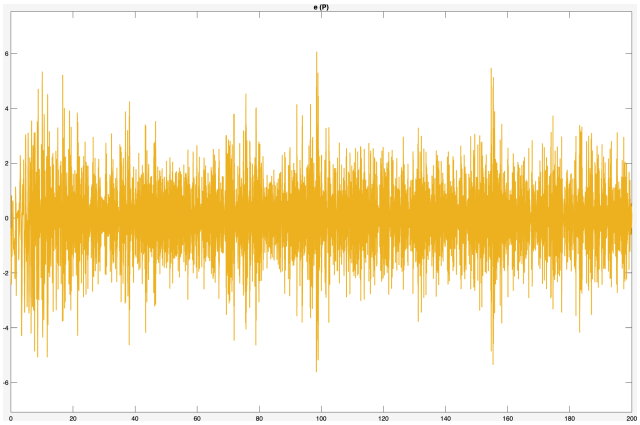


8.1 Measurement noise

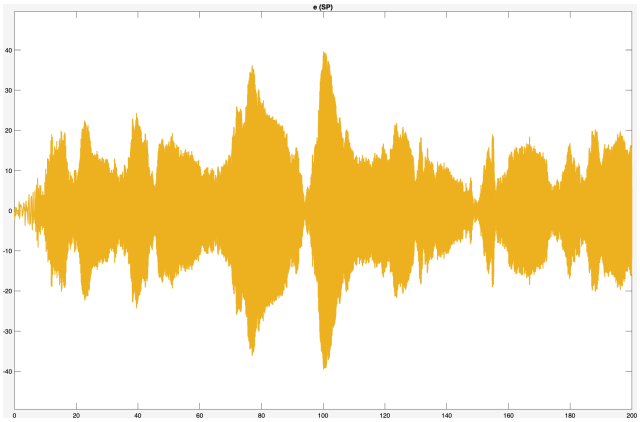
8.1.1 State estimation error $e = \hat{x} - x$

With noise in the state measurement x, the results are as follows.

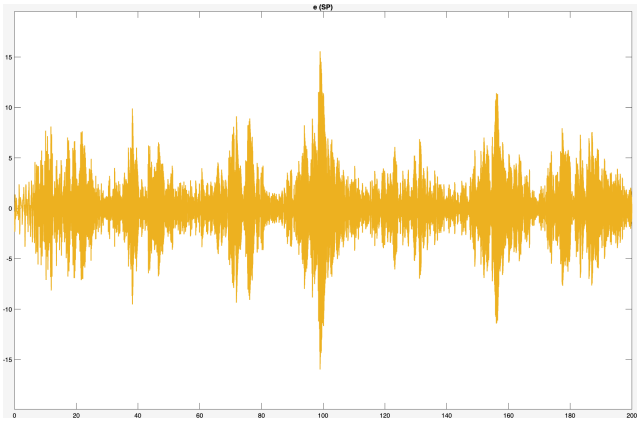
- In the parallel model, the estimation error e fluctuates within ± 6.



- In the series-parallel model with $a_m = 0.1$, the estimation error e fluctuates within ± 40.



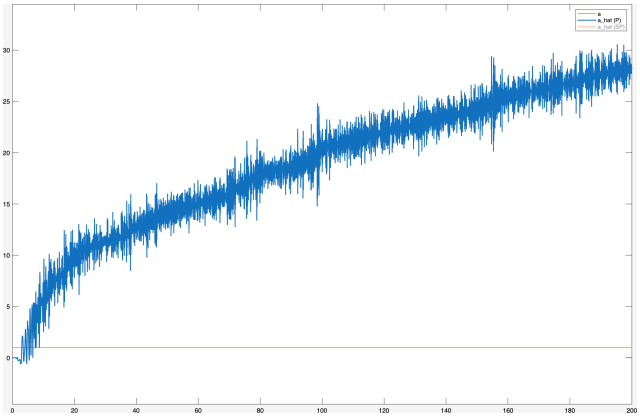
- In the series-parallel model with $a_m = 2$, the estimation error e fluctuates within ± 16.



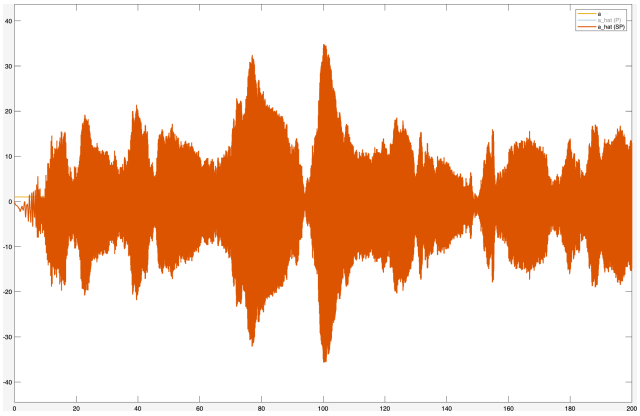
8.1.2 Estimation error for a

With noise in the state measurement x, the results are as follows.

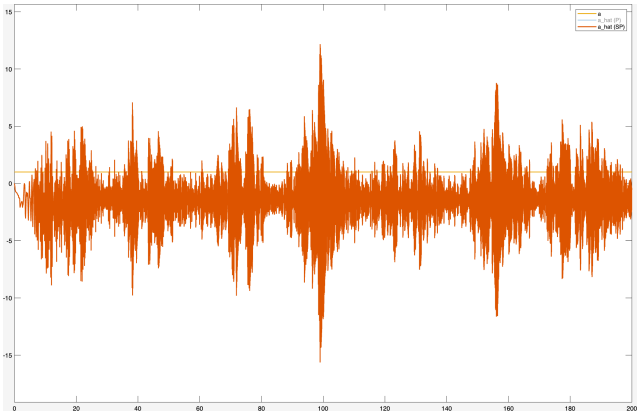
- In the parallel model, the estimation error for a $\hat{a}$ diverges.



- In the series-parallel model with $a_m = 0.1$, the estimation error for a $\hat{a}$ fluctuates within ± 40.



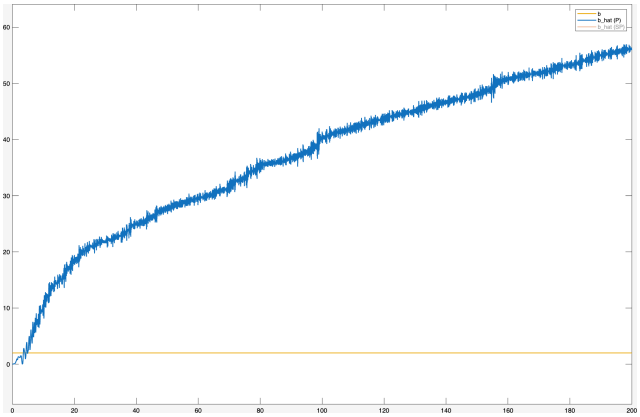
- In the series-parallel model with $a_m = 2$, the estimation error for a $\hat{a}$ fluctuates between -18 and $+14$.



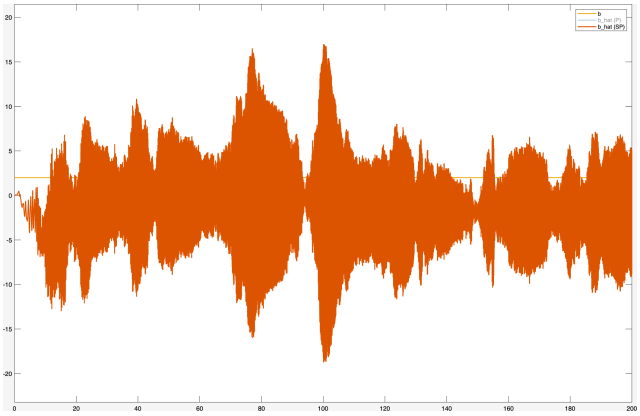
8.1.3 Estimation error for b

With noise in the state measurement x, the results are as follows.

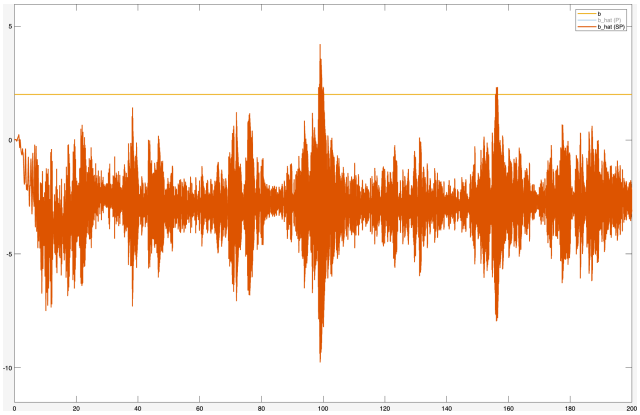
- In the parallel model, the estimation error for b $\hat{b}$ diverges.



- In the series-parallel model with $a_m = 0.1$, the estimation error for b $\hat{b}$ fluctuates within ± 20.



- In the series-parallel model with $a_m = 2$, the estimation error for b $\hat{b}$ fluctuates between -10 and $+5$.

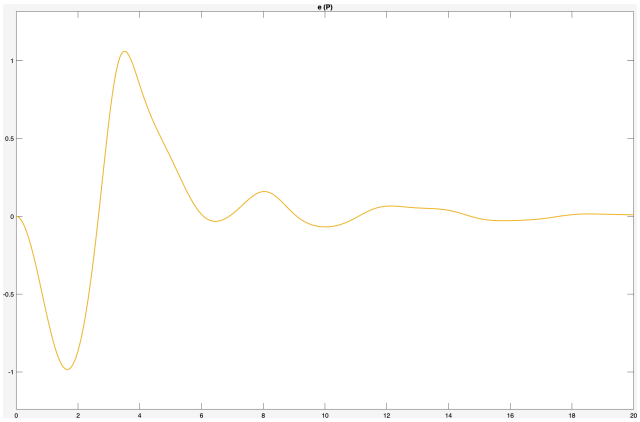


9 Parameter $a = 1$, sinusoidal input $u = \sin t$

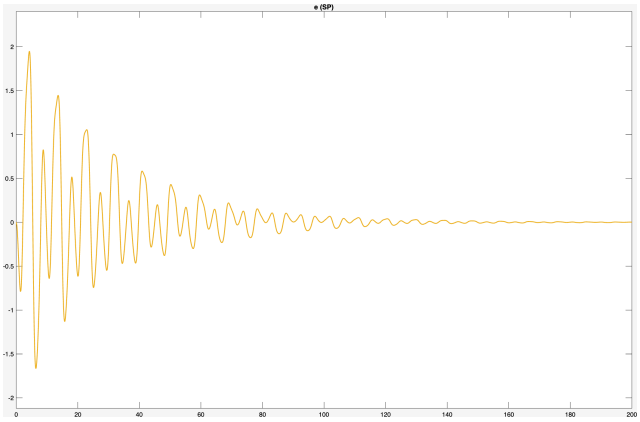
9.0.1 State estimation error $e = \hat{x} - x$

The results are as follows.

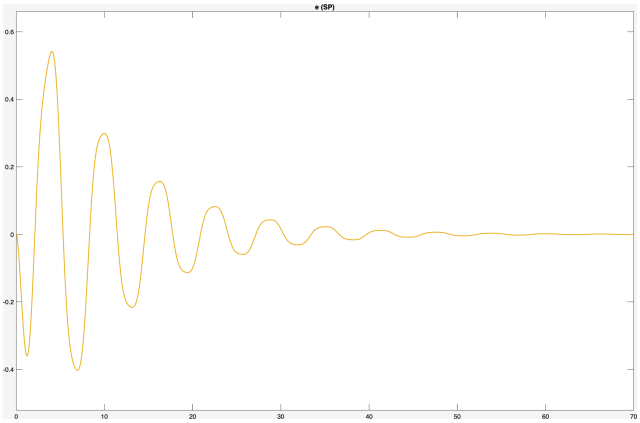
- In the parallel model, the estimation error e converges in approximately $19s$.



- In the series-parallel model with $a_m = 0.1$, the estimation error e converges in approximately $190s$.



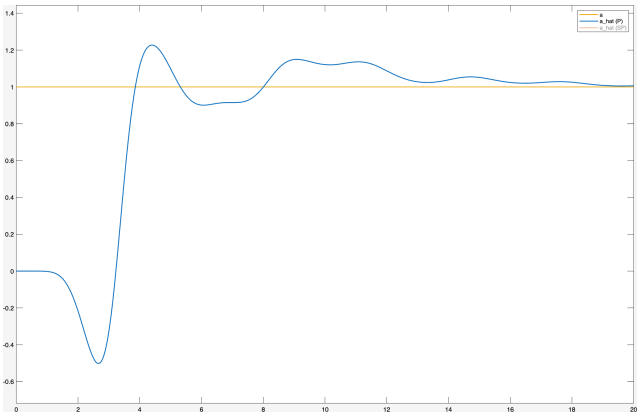
- In the series-parallel model with $a_m = 0.1$, the estimation error e converges in approximately $65s$.



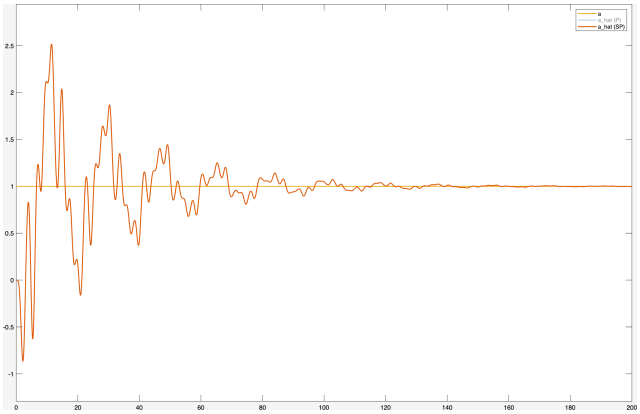
9.0.2 Estimation error for a

The results are as follows.

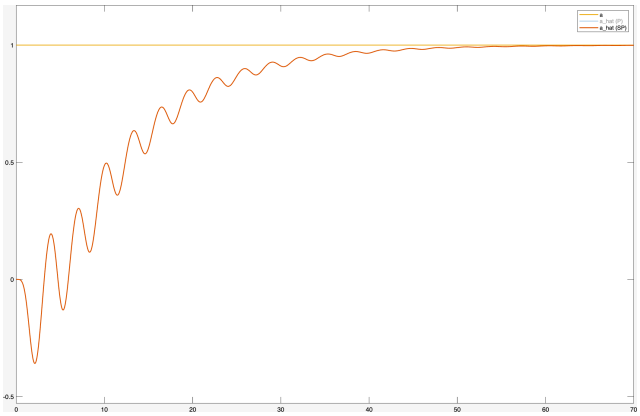
- In the parallel model, the estimation error for a $\hat{a}$ converges in approximately $19s$ to the true value of a.



- In the series-parallel model with $a_m = 0.1$, the estimation error for a $\hat{a}$ converges in approximately $170s$ to the true value of a.



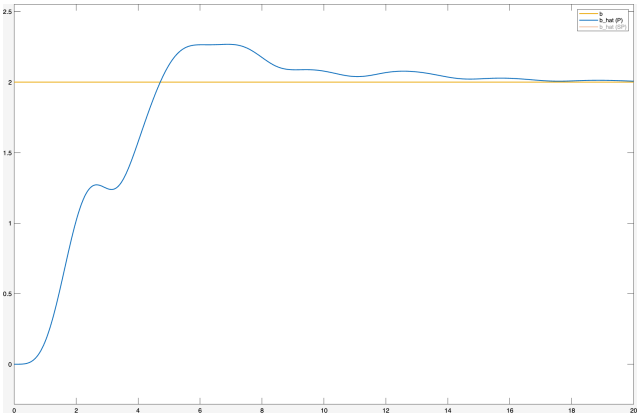
- In the series-parallel model with $a_m = 2$, the estimation error for a $\hat{a}$ converges in approximately $55s$ to the true value of a.



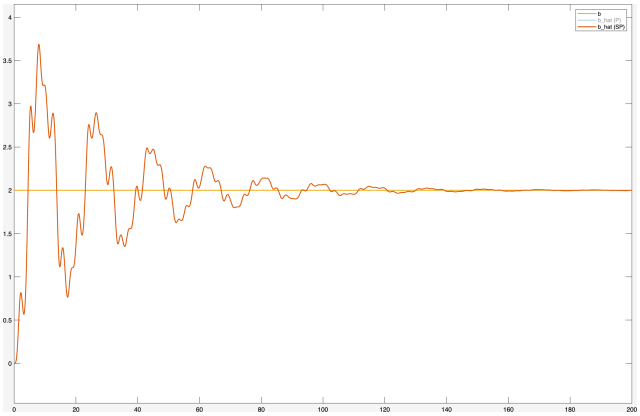
9.0.3 Estimation error for b

The results are as follows.

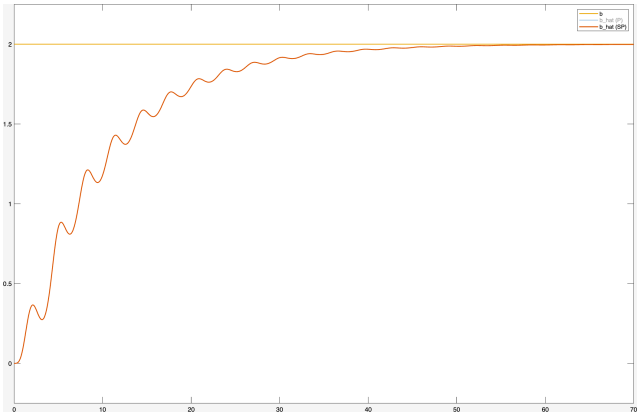
- In the parallel model, the estimation error for b, $\hat{b}$, converges in approximately $20s$ to 2, the true value of b.



- In the series-parallel model with $a_m = 0.1$, the estimation error for b $\hat{b}$ converges in approximately $170s$ to the true value of b.



- In the series-parallel model with $a_m = 0.1$, the estimation error for b $\hat{b}$ converges in approximately $60s$ to the true value of b.

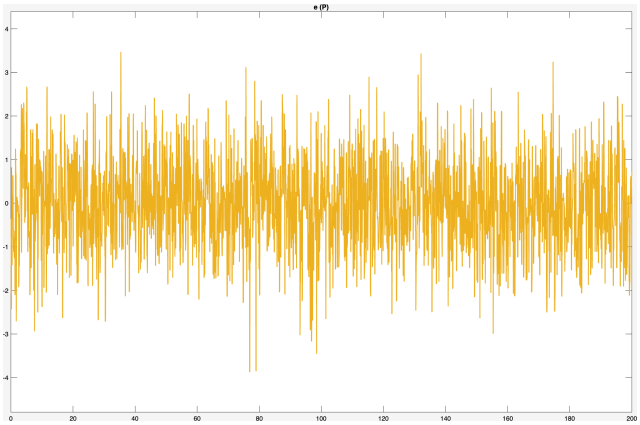


9.1 Measurement noise

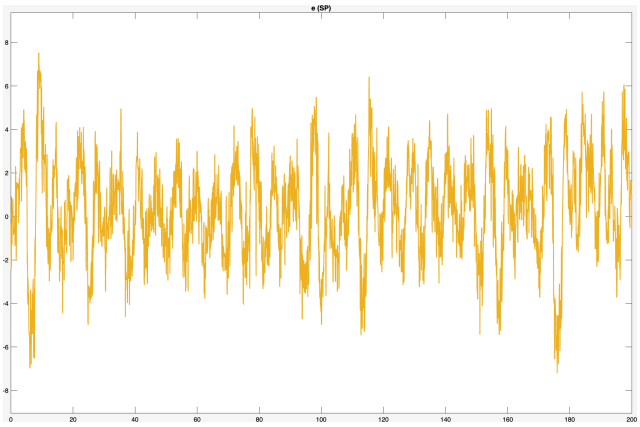
9.1.1 State estimation error $e = \hat{x} - x$

With noise in the state measurement x, the results are as follows.

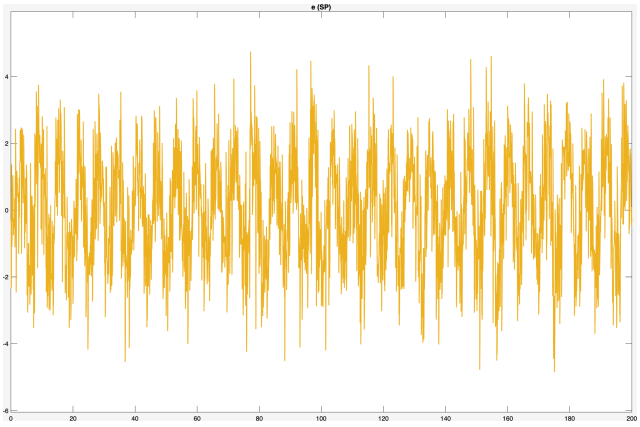
- In the parallel model, the estimation error e fluctuates within ± 4.



- In the series-parallel model with $a_m = 0.1$, the estimation error e fluctuates within ± 8.



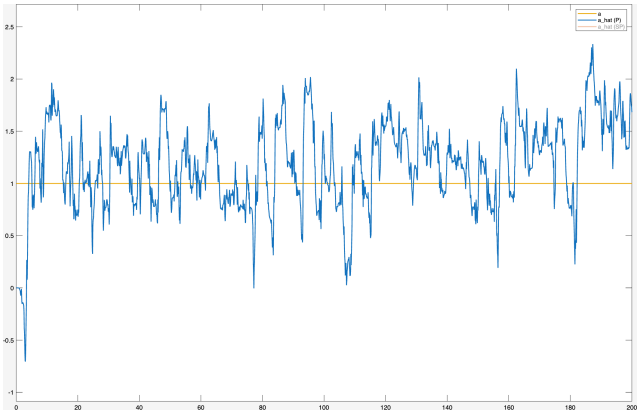
- In the series-parallel model with $a_m = 2$, the estimation error e fluctuates within ± 5.



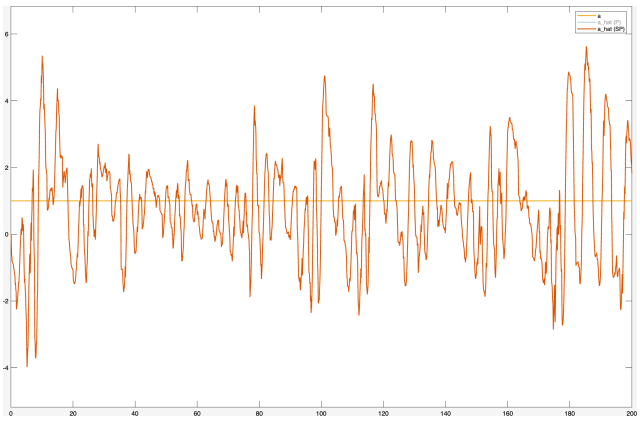
9.1.2 Estimation error for a

With noise in the state measurement x, the results are as follows.

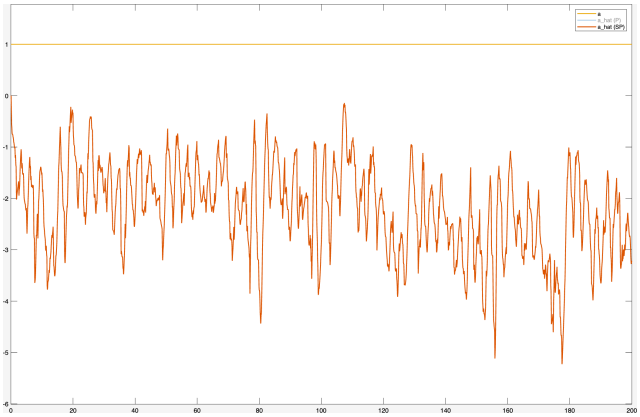
- In the parallel model, the estimation error for a $\hat{a}$ fluctuates between -1 and 2.5.



- In the series-parallel model with $a_m = 0.1$, the estimation error for a $\hat{a}$ fluctuates between -4 and $+6$.



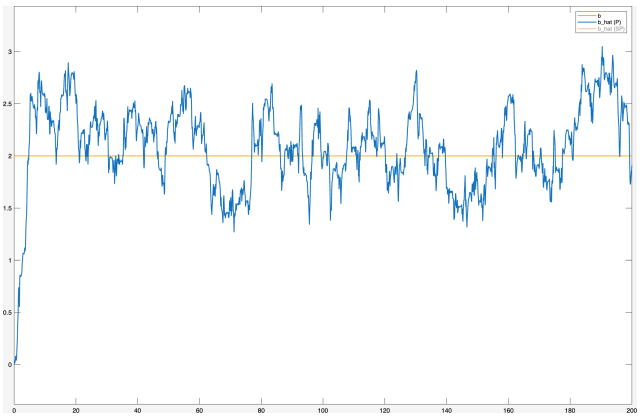
- In the series-parallel model with $a_m = 2$, the estimation error for a $\hat{a}$ fluctuates between -6 and 0.



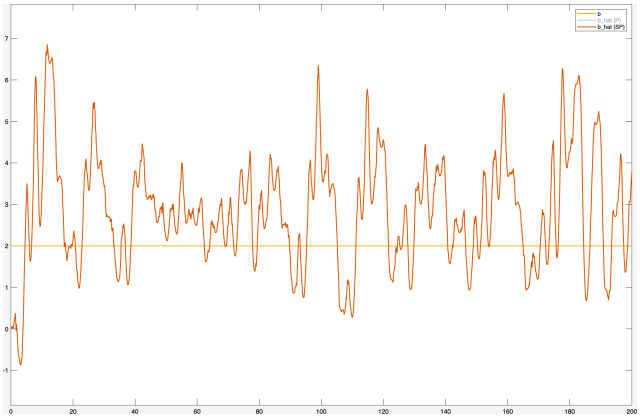
9.1.3 Estimation error for b

With noise in the state measurement x, the results are as follows.

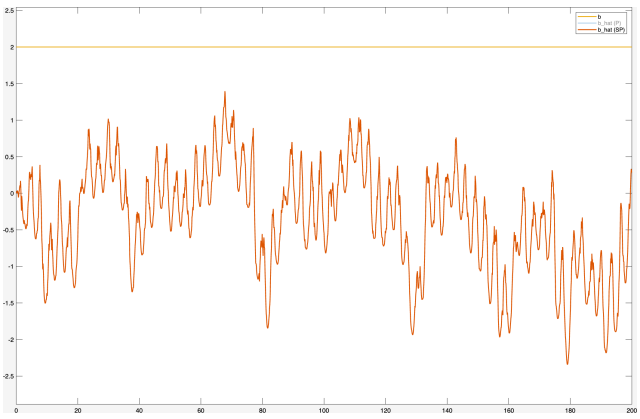
- In the parallel model, the estimation error for b $\hat{b}$ fluctuates between 0 and 3.



- In the series-parallel model with $a_m = 0.1$, the estimation error for b $\hat{b}$ fluctuates between -1 and $+7$.



- In the series-parallel model with $a_m = 2$, the estimation error for b $\hat{b}$ fluctuates between -2.5 and 1.5.

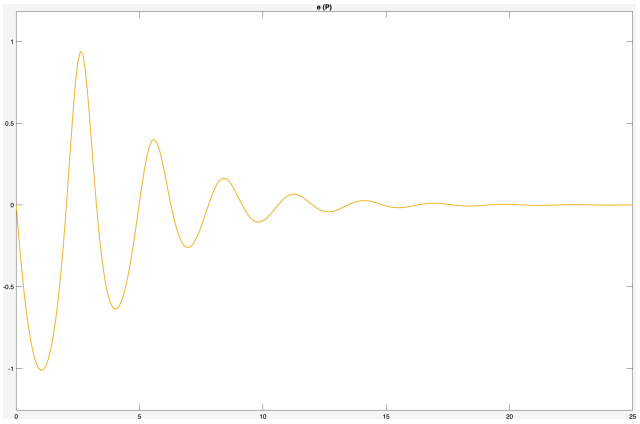


10 Parameter $a = 1 + \sin\left(\frac{2\pi t}{24 \times 3600}\right)$, constant input $u = 1$

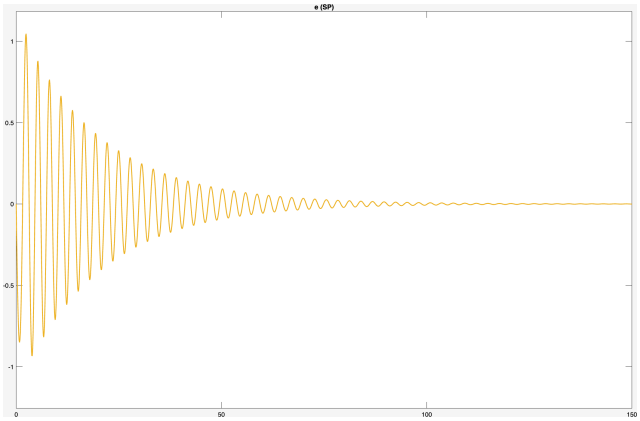
10.0.1 State estimation error $e = \hat{x} - x$

The results are as follows.

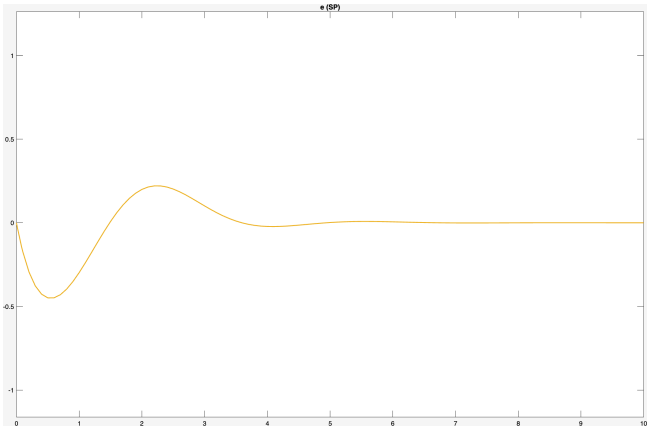
- In the parallel model, the estimation error e converges in approximately $23s$.



- In the series-parallel model with $a_m = 0.1$, the estimation error e converges in approximately $140s$.



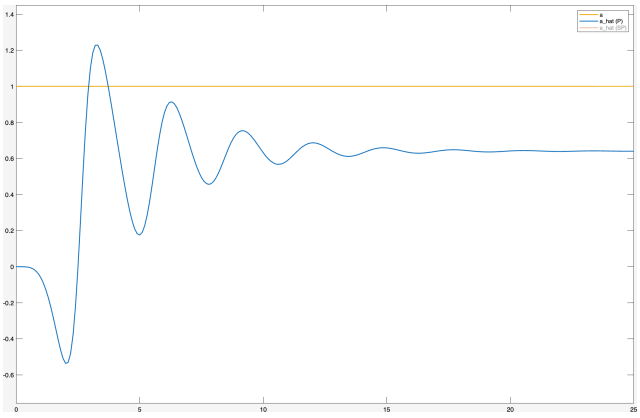
- In the series-parallel model with $a_m = 0.1$, the estimation error e converges in approximately $7s$.



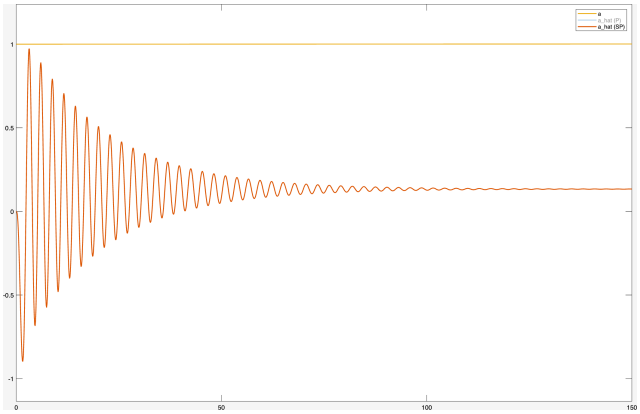
10.0.2 Estimation error for a

The results are as follows.

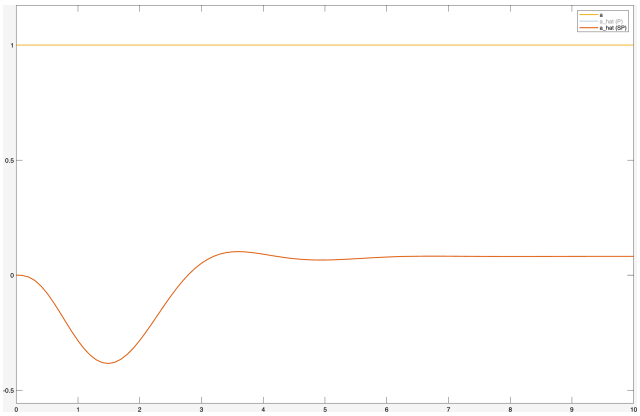
- In the parallel model, the estimation error for a $\hat{a}$ converges in approximately $22s$ to 0.64.



- In the series-parallel model with $a_m = 0.1$, the estimation error for a $\hat{a}$ converges in approximately $130s$ to 0.13.



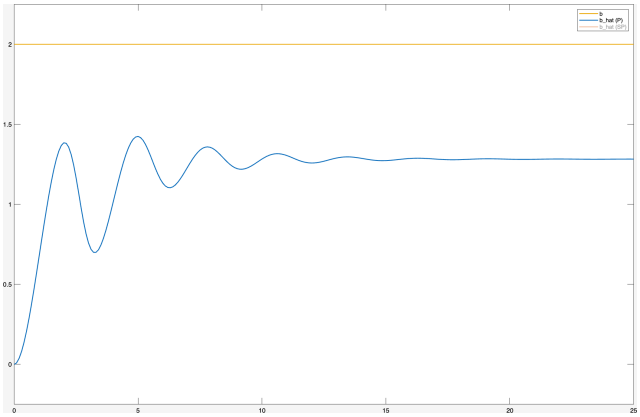
- In the series-parallel model with $a_m = 0.1$, the estimation error for a $\hat{a}$ converges in approximately $8s$ to 0.08.



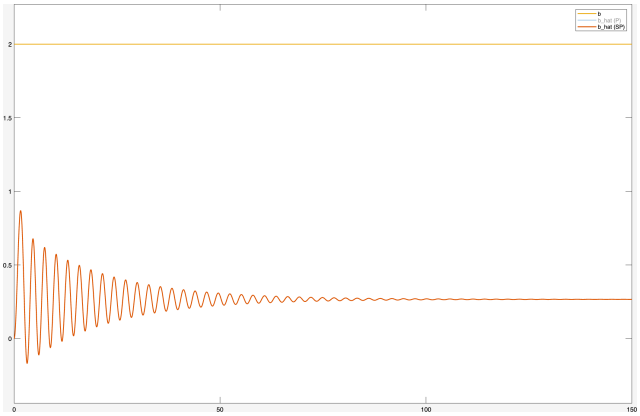
10.0.3 Estimation error for b

The results are as follows.

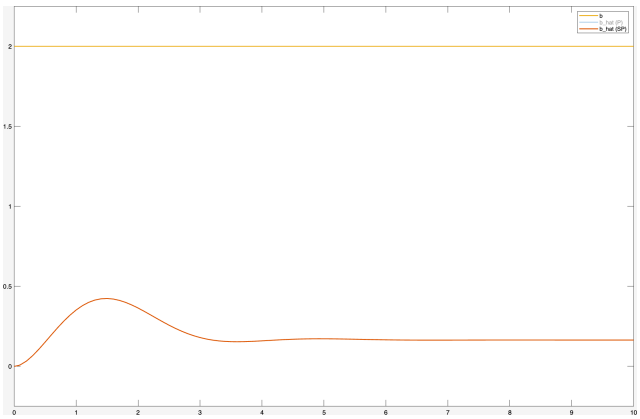
- In the parallel model, the estimation error for b $\hat{b}$ converges in approximately $22s$ to 1.28.



- In the series-parallel model with $a_m = 0.1$, the estimation error for b $\hat{b}$ converges in approximately 140 s to 0.27.



- In the series-parallel model with $a_m = 0.1$, the estimation error for b $\hat{b}$ converges in approximately $6s$ to 0.16.

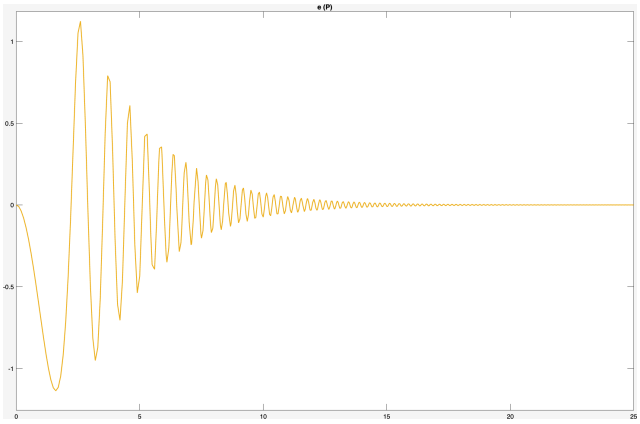


11 Parameter $a = 1 + \sin\left(\frac{2\pi t}{24 \times 3600}\right)$, ramp input $u = t$

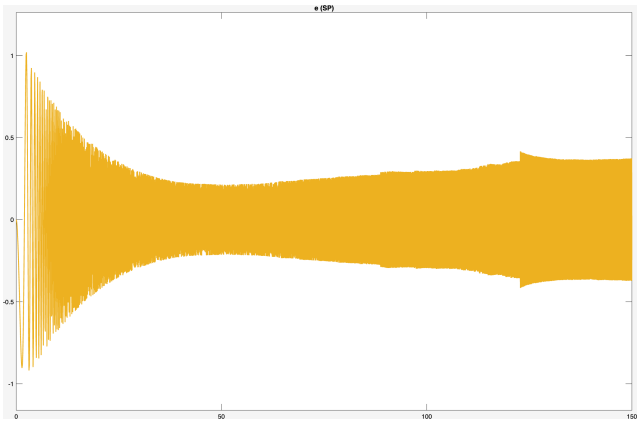
11.0.1 State estimation error $e = \hat{x} - x$

The results are as follows.

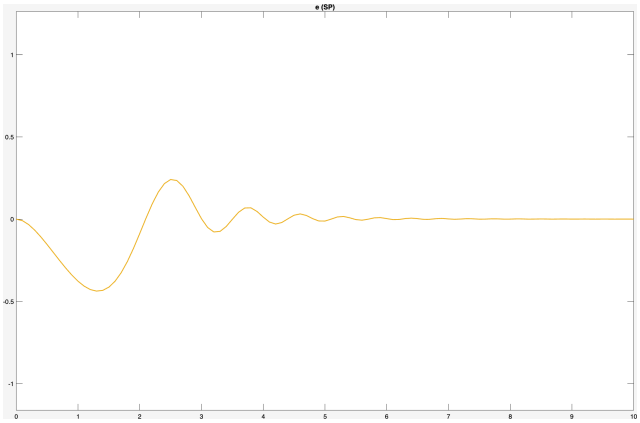
- In the parallel model, the estimation error e converges in approximately $22s$.



- In the series-parallel model with $a_m = 0.1$, the estimation error e diverges.



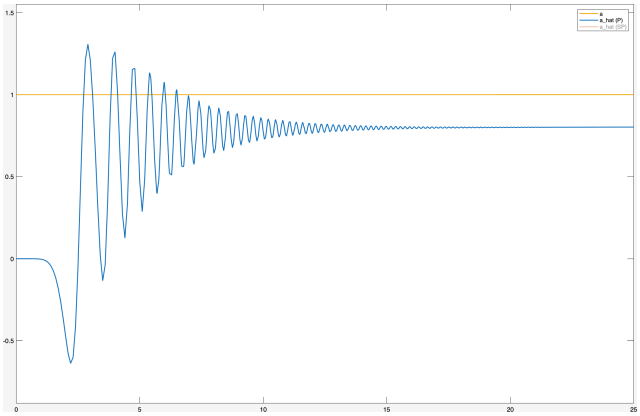
- In the series-parallel model with $a_m = 2$, the estimation error e converges in approximately $7s$.



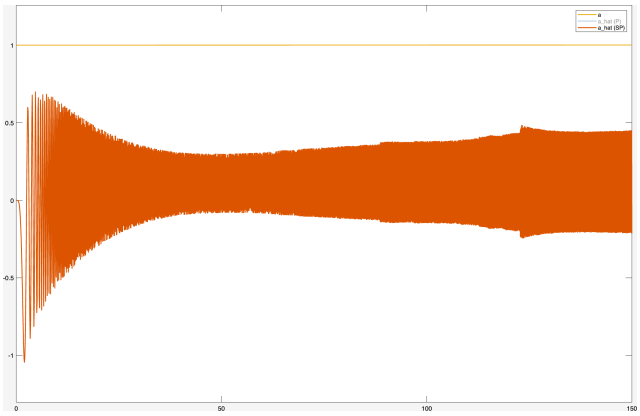
11.0.2 Estimation error for a

The results are as follows.

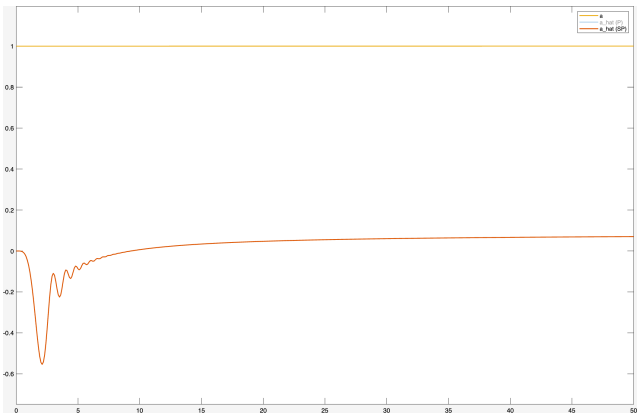
- In the parallel model, the estimation error for a $\hat{a}$ converges in approximately $23s$ to 0.8.



- In the series-parallel model with $a_m = 0.1$, the estimation error for a $\hat{a}$ diverges.



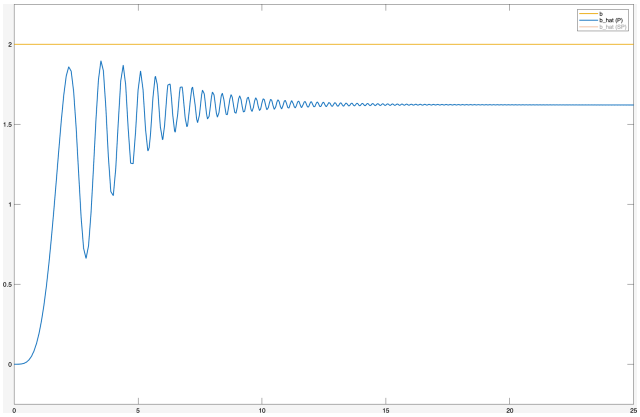
- In the series-parallel model with $a_m = 2$, the estimation error for a $\hat{a}$ converges in approximately $44s$ to 0.07.



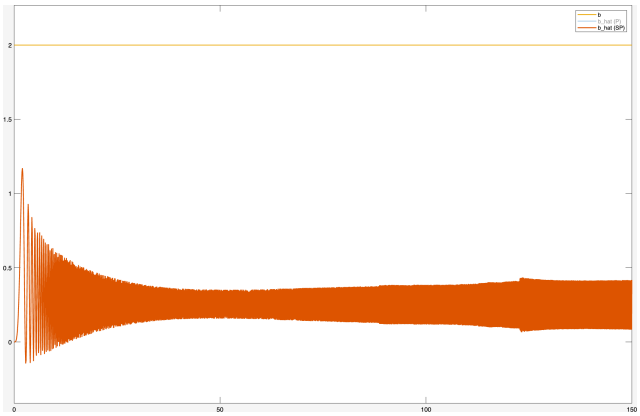
11.0.3 Estimation error for b

The results are as follows.

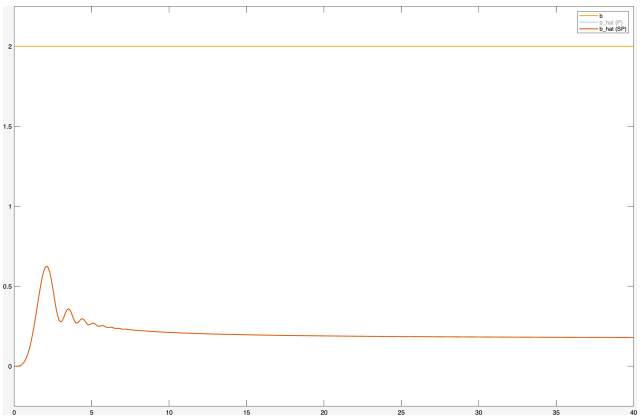
- In the parallel model, the estimation error for b $\hat{b}$ converges in approximately $20s$ to 1.62.



- In the series-parallel model with $a_m = 0.1$, the estimation error for b $\hat{b}$ diverges.



- In the series-parallel model with $a_m = 0.1$, the estimation error for b $\hat{b}$ converges in approximately $36s$ to 0.18.

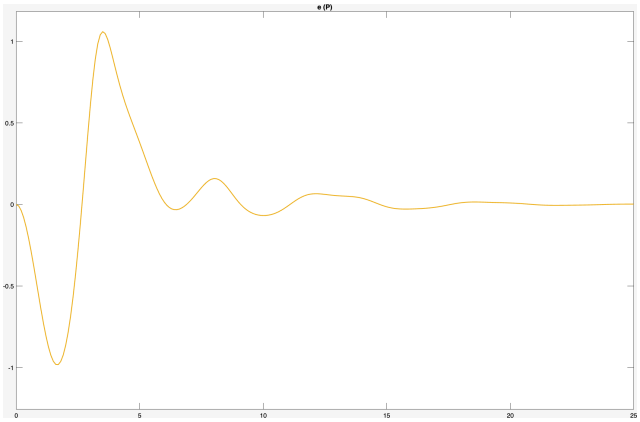


12 Parameter $a = 1 + \sin\left(\frac{2\pi t}{24 \times 3600}\right)$, sinusoidal input $u = \sin t$

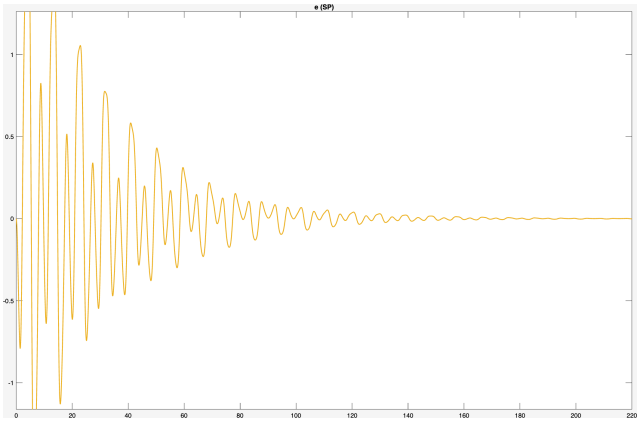
12.0.1 State estimation error $e = \hat{x} - x$

The results are as follows.

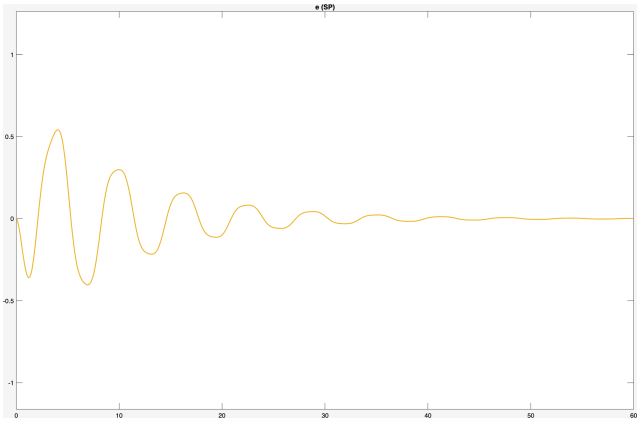
- In the parallel model, the estimation error e converges in approximately $22s$.



- In the series-parallel model with $a_m = 0.1$, the estimation error e converges in approximately $210s$.



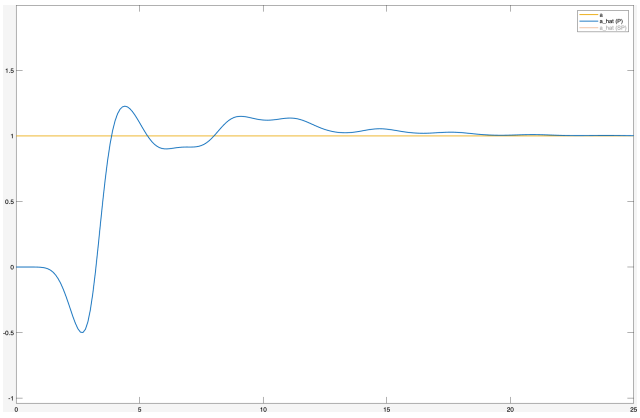
- In the series-parallel model with $a_m = 2$, the estimation error e converges in approximately $56s$.



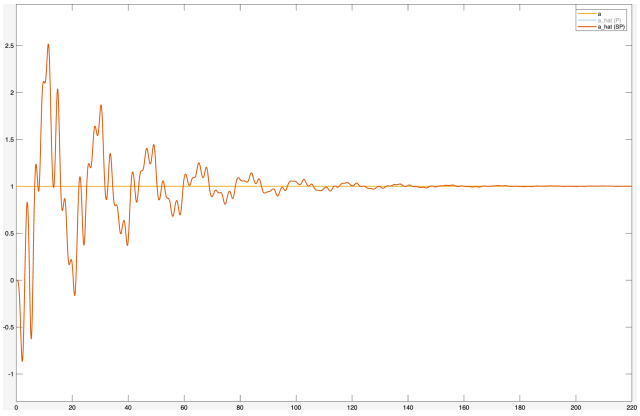
12.0.2 Estimation error for a

The results are as follows.

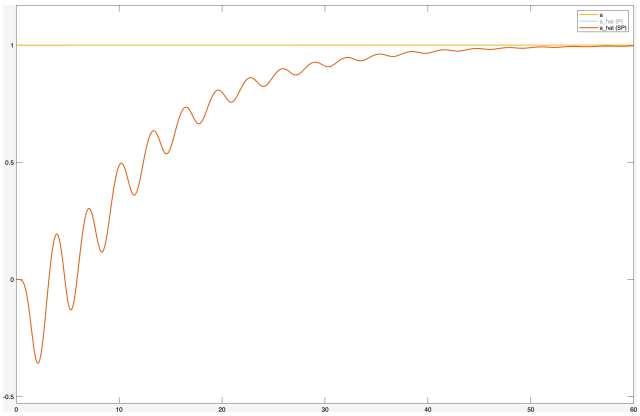
- In the parallel model, the estimation error for a $\hat{a}$ converges in approximately $22s$ to the true value of a.



- In the series-parallel model with $a_m = 0.1$, the estimation error for a $\hat{a}$ converges in approximately $200s$ to the true value of a.



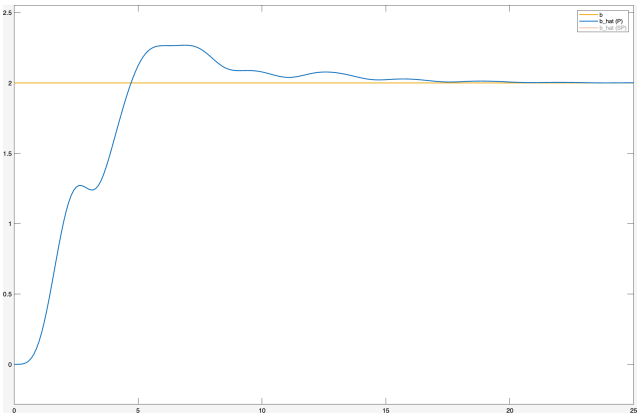
- In the series-parallel model with $a_m = 2$, the estimation error for a $\hat{a}$ converges in approximately $60s$ to the true value of a.



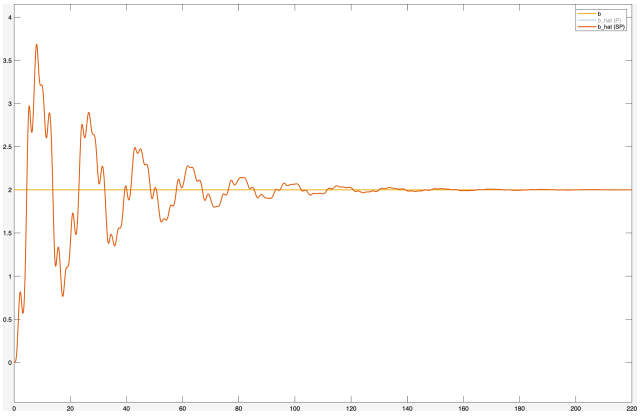
12.0.3 Estimation error for b

The results are as follows.

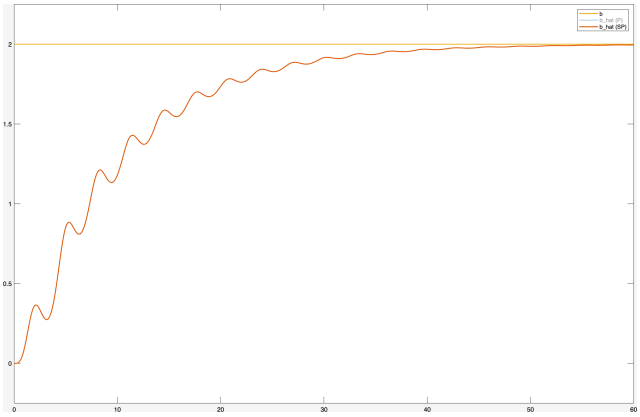
- In the parallel model, the estimation error for b $\hat{b}$ converges in approximately $22s$ to the true value of b.



- In the series-parallel model with $a_m = 0.1$, the estimation error for b $\hat{b}$ converges in approximately $210s$ to the true value of b.



- In the series-parallel model with $a_m = 2$, the estimation error for b $\hat{b}$ converges in approximately $55s$ to the true value of b.



13 Conclusions

The simulations yielded the following observations.

- **In the series-parallel model, increasing a_m makes the state estimation error e and the parameter estimates $\hat{a}$ and $\hat{b}$ converge faster**.

  The error dynamics

  $$\dot{e} \;=\; -a_m\, e - \tilde{a}x + \tilde{b}u$$

  have a pole at $s = -a_m$, so increasing a_m makes the error dynamics faster. The adaptation laws

  $$\begin{aligned} \dot{\hat{a}} &= e\, x \\ \dot{\hat{b}} &= -e\, u \end{aligned}$$

  depend directly on the error e, so the parameter estimates generally converge faster as well.

- **With a constant or ramp input, the parameter estimates $\hat{a}$ and $\hat{b}$ from either model fail to converge to the true values of a and b. With a sinusoidal input, both models identify the two parameters correctly.**

  This occurs because the sinusoidal input is persistently exciting, unlike the constant and step inputs. For a ramp input, error convergence depends on the model parameters chosen in the simulation.

- **In the cases studied, the parallel model estimated the parameters more accurately.** This behaviour depends on the simulation parameters and can vary between cases.

- **When the state measurement is noisy, the estimation error e and the parameter estimates $\hat{a}$ and $\hat{b}$ fail to converge.**

  If the state measurement is affected by noise $n(t)$, the measured state is given by

  $$x_m \;=\; x + n$$

  and the measurement error used in the adaptation laws is

  $$\begin{aligned} e_m &= \hat{x} - x_m \\ &= e - n \end{aligned}$$

  Consequently,

  $$\dot{e}_m \;=\; \dot{e} - \dot{n}$$

  The derivative of the noise $\dot{n}$ can be unbounded, so $\dot{e}_m \in L_\infty$ is not guaranteed. As a result, the estimation error e may fail to converge asymptotically to zero, and the parameter estimates $\hat{a}$ and $\hat{b}$ may also fail to converge.

- **With state measurement noise, the parallel model is more robust than the series-parallel model**.

Recall that the parallel model is described by

$$\begin{aligned}\dot{\hat{x}} &= -\hat{a}\hat{x} + \hat{b}u \\ \dot{\hat{a}} &= e_m\hat{x} \\ \dot{\hat{b}} &= -e_m u\end{aligned}$$

while the series-parallel model is described by

$$\begin{aligned}\dot{\hat{x}} &= -a_m\hat{x} + (a_m - \hat{a})x_m + \hat{b}u \\ \dot{\hat{a}} &= e_m x_m \\ \dot{\hat{b}} &= -e_m u\end{aligned}$$

In the parallel model, the noise n directly affects only the calculation of the error e_m. In the series-parallel model, it affects the calculation of e_m, the dynamics of $\hat{x}$, and the dynamics of $\hat{a}$ and $\hat{b}$. The noise therefore enters more equations in the series-parallel model, making it generally more sensitive to measurement disturbances.

- **With state measurement noise, a larger a_m in the series-parallel model generally produces a smaller state estimation error**.

  This result depends on the system parameters and the characteristics of the noise. Given

  $$\dot{e}_m = -a_m e_m - \tilde{a}x_m + \tilde{b}u$$

  a larger value of a_m causes the measurement error to decay faster.

  At the same time, the state estimate dynamics

  $$\dot{\hat{x}} = -a_m\hat{x} + (a_m - \hat{a})x_m + \hat{b}u$$

  are more strongly affected by very large values of a_m.

  In the cases analysed, the effect of faster error decay outweighed the influence of a_m on the dynamics of $\hat{x}$.

- **When the parameter a varies over time, the estimation error e and the estimation errors for $\hat{a}$ and $\hat{b}$ converge with step or sinusoidal inputs and diverge with a ramp input.**

  However, the theory does not guarantee error convergence because

  $$\tilde{a} = \hat{a} - a(t)$$

  and therefore, in the parallel model,

  $$\begin{aligned}\dot{S} &= \dot{S}_1 + \dot{S}_2 \\ &= e\,\dot{e} + \tilde{a}\,\dot{\tilde{a}} + \tilde{b}\dot{\tilde{b}} \\ &= e(-a\,e - \tilde{a}\hat{x} + \tilde{b}u) + \tilde{a}(\dot{\hat{a}} - \dot{a}(t)) + \tilde{b}\dot{\tilde{b}} \\ &= -a\,e^2 - \cancel{\tilde{a}\hat{x}}\,e + \tilde{b}\cancel{u}\,e + \tilde{a}\cancel{e}\,\hat{x} - \tilde{a}\,\dot{a}(t) - \tilde{b}\cancel{e}\,u \\ &= -a\,e^2 - \tilde{a}\,\dot{a}(t)\end{aligned}$$

while in the series-parallel model,

$$\begin{aligned}
\dot{S} &= \dot{S}_1 + \dot{S}_2 \\
&= e\,\dot{e} + \tilde{a}\,\dot{\tilde{a}} + \tilde{b}\dot{\tilde{b}} \\
&= e(-a_m\,e - \tilde{a}x + \tilde{b}u) + \tilde{a}(\dot{\hat{a}} - \dot{a}(t)) + \tilde{b}\dot{\hat{b}} \\
&= -a_m\,e^2 - \cancel{\tilde{a}x}\,e + \cancel{\tilde{b}u}\,e + \cancel{\tilde{a}x}\,e - \tilde{a}\,\dot{a}(t) - \cancel{\tilde{b}u}\,e \\
&= -a_m\,e^2 - \tilde{a}\,\dot{a}(t)
\end{aligned}$$

The storage function S therefore cannot be used as a Lyapunov function to prove $\hat{a}, \hat{b}, e \in L_\infty$ and, consequently, $e(t) \to \infty$.

Convergence therefore depends on the parameters used in the model.